

Lecture 1. The Research Process

Part 1. Module outline

Weekly activities

Mon	Tues	Wed	Thurs	Fri
Required reading				Read Q&A answers
	Self-test quizzes			
	9am Drop-ins			a.m. Practical class
	1pm Lecture			
6pm Peer mentors		Post questions before 5pm		
	Midnight Homework deadline			

-  = compulsory classes
-  = optional support sessions
-  = independent work

Asking questions

- **“Hot Questions”** on Moodle: post any questions you might have on the lecture content before 5pm on Wednesday.
- **Peer discussion forum:** monitored by your peer mentors.
- **Staff forum:** monitored by your practical demonstrators.
- **Coding clinics:** weekly online appointments you can book with the demonstrators every Tuesday at 9-10. One slot is 15 minutes but you can double-book for a longer chat.
- **Peer mentors:** optional weekly group meetings with your peer mentors (Year 2 students) – just pop by and say hello.
- On coding: **Google** (no, seriously!)

Assessment

Component	Time	Weight
Homework assignments	Tuesday midnight	10%
Multiple choice test	12 th December 2025	10%
Final exam	6 th March 2026	80%

Homework

- Weekly assignment due **midnight** on **Tuesday** each week
- Complete the digital worksheet and upload it to Moodle + Fill in the reflections form
- Credit is given for completing the work not for accuracy – so if you submit all 13 assignments you will get a mark of 100 for the homework component, which is scaled down to 10% of the module mark.
- “Complete” means you have attempted (almost) all of the questions.

Exam

- Exams are all in person rather than online this year
- All assignments are Open Book – this means you can access the internet and your notes during the exam. Note, however, that you cannot use generative AI like ChatGPT in any way.
- You **must not** communicate with others during the exam. This includes using shared Google documents.

Part 2. Why study statistics and R?

People have always had theories about the human mind. What makes modern science unique is that for the first time in history, people are basing their theories on **empirical research** (empirical = derived from real-world observations of data) rather than on intuition or pure reasoning.

Empirical research can take on many forms, but we can divide them into two broad categories:

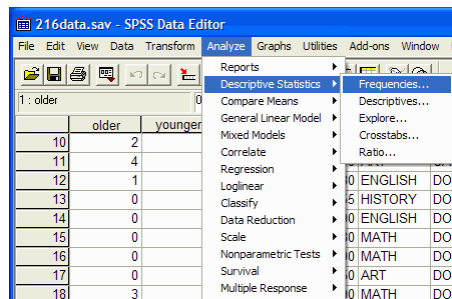
- **Qualitative** research: involves collecting data that cannot be quantified, such as:
 - interviews about people’s experiences, beliefs, and life narratives;
 - focus groups (group discussions revealing how people debate and navigate a topic);
 - other textual (e.g., diaries) or non-textual data (e.g., media analysis).

- **Quantitative** research: involves quantifiable data (questionnaire scores; reaction times; number of occurrences of a particular event; performance on a task, etc.)

Why can't we use individual observations / clinical insights?

Why use R?

Until a couple of years ago, we used to use a programme called SPSS for data analysis – you could run statistical tests (e.g., comparing scores between groups) by choosing the right test from a drop-down menu.



Instead of drop-down menus, R uses code – you write out the operations that you want to perform on your data. This has a number of benefits:

- R allows us to create **reproducible** reports – you can share your data and the code for the analysis with other researchers, allowing them to repeat your analysis.
- Coding is a transferable skill that is valued on the job market.

Statistics / coding anxiety

Maths and coding are NOT innate, the majority of skill level is determined by practice.

Some of you will be anxious about taking a course in statistics and coding – this is normal.

It's important that you complete the practical and homework. Each session builds on the skills you gained in previous sessions.

Part 3. The Research Process

Step 1. Choose a research question

Observation: I find it hard to remember things when I am tired
⇒ Research question: Does sleep impact memory?

Step 2. Select dependent and independent variables

Independent variable (IV, **predictor**) = “the thing doing the explaining”

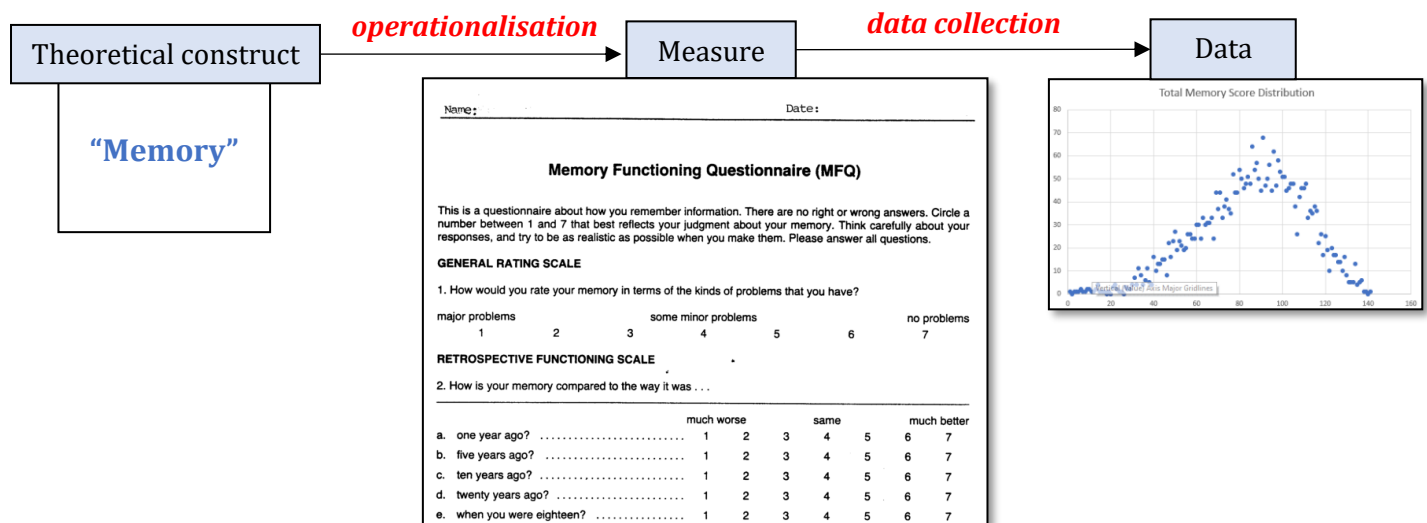
Dependent variable (DV, **outcome**) = “the thing being explained”

Research question: Does sleep impact memory?

⇒ IV = sleep, DV = memory

Step 3. Operationalise the variables

Operationalisation = transforming an abstract, theoretical construct (like “memory”) into something concrete and measurable (such as the number of words recalled in a memorisation task)



There is more than one way of operationalising your variables!

For example, sleep:

- Do we measure sleep *quality*? Or *duration*?
- How precise to we need the measurement to be? Hours, minutes, seconds?
- How do we gather this information? Self-report? Observation? Electrophysiology?

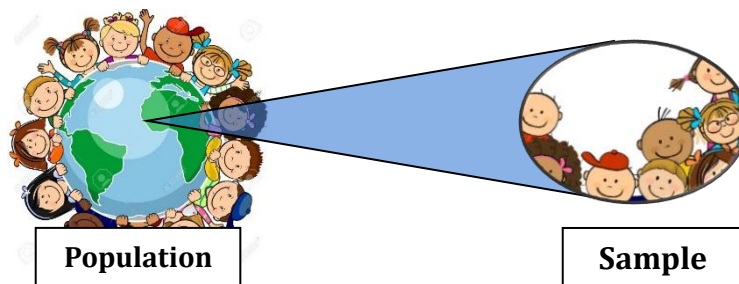
Step 4. State your hypothesis

A **hypothesis** is a testable belief about the outcome of a study. It should be based on a theory or previous research and formulated in terms of your operationalised variables:

"People that report having better quality sleep will score higher on the memory test"

Step 5. Select a sample from the population of interest

- **Population** = the group people you are interested in studying.
- There are usually way more people than you are able to collect data from, so you select a **sample** from the population
- Your sample should be **representative** of the population



Step 6. Collect data

- AND put it in a usable format
- Identify errors / missing data
- Sum items on a questionnaire to give a total score (e.g., single measure of sleep quality)

or_q10eal	q10mainl	q11school	gonogoco	gonogoun	gonogoatt	gonogoex	gonogo_t	gonogo_a	gonogo_ci	rt_trials	scort_acc
Monolingual	No	Yes	5	5	No	110	32	1	30	24	
Monolingual	Yes	Yes	4	5	No	110	22	1	30	26	
Monolingual	No	Yes	3	4	No	110	17	1	30	22	
Monolingual	No	Yes	5	5	No	110	30	1	30	29	
Monolingual	No	Yes	3	5	No	110	24	1	30	23	
Monolingual	Yes	Yes	4	3	No	110	14	1	30	27	
Monolingual	No	Yes	5	5	No	110	29	1	30	24	
Monolingual	Yes	Yes	3	5	No	110	36	1	30	20	
Monolingual	No	Yes	5	3	No	110	25	1	30	24	
Monolingual	No	Yes	3	4	No	110	36	1	30	27	
Monolingual	No	Yes	3	4	No	110	18	1	30	22	
Monolingual	No	Yes	5	3	No	110	29	1	30	27	
Monolingual	Yes	Yes	3	4	No	110	20	1	30	21	
Monolingual	No	Yes	3	4	No	110	16	1	30	28	
Monolingual	Yes	Yes	4	5	No	110	34	1	30	21	
Monolingual	No	Yes	4	4	No	110	6	1	30	30	
Monolingual	No	Yes	4	4	No	110	38	1	30	28	

Step 7. Analyse data

- Summarise (describe) data with **descriptive statistics**: minimum and maximum scores, averages, etc.
- Visualise the data in graphs
- Run **inferential statistical tests** to determine whether the data support our hypothesis

Step 8. Draw conclusions

- State whether your data support your hypothesis or not
- No one study can support a hypothesis conclusively
 - Our measures may not map on well to our theoretical construct
 - Our sample may not be representative
- Revise theory and identify next research question